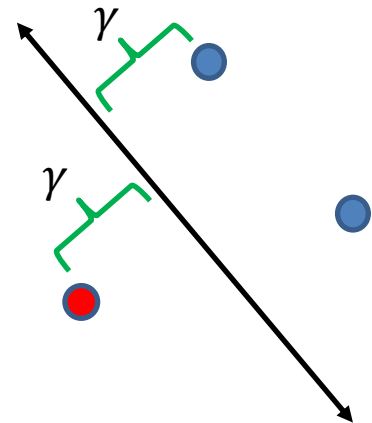


Support Vector Machines

Lee-Ad Gottlieb

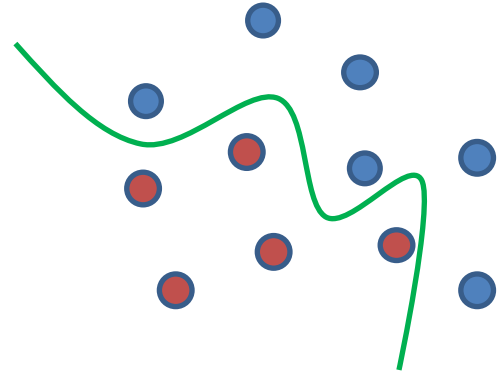
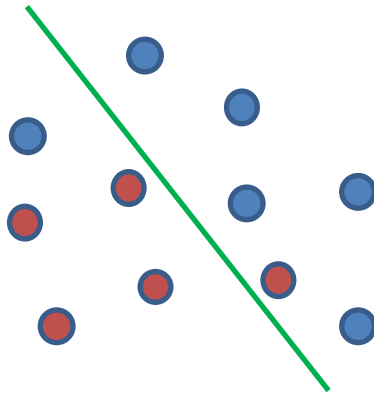
Separating hyperplanes

- Review: We saw several algorithms to find a separating hyperplane
 - Brute force
 - Winnow
 - Perceptron
 - Clarkson's algorithms
- What is the VC-dimension of separating hyperplanes?
 - d dimensions: $d+1$ Could be large
 - Margin γ : $\frac{O(1)}{\gamma^2}$



Overfitting

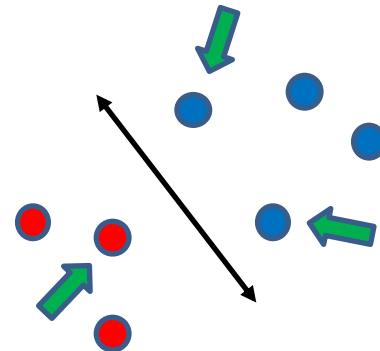
- Why is high VC-dimension bad
 - Like memorizing...



- **Overfitting**
 - Learning the sample, not understanding the underlying rule behind the sample

Hard-margin SVM

- Suppose the data is separable
 - We want to find a hyperplane separating the points with large margin
 - A **hard margin** is consistent with S
- A solution was given by Vapnik (with others) in 1990's, while he was at Bell Labs, AT&T.
- **Support Vector Machines**

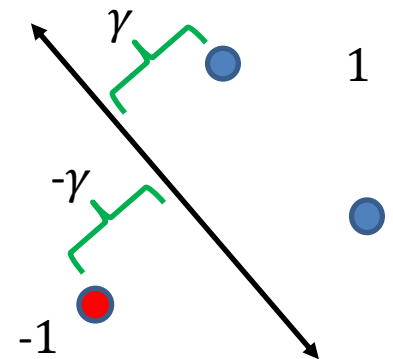


Hard-margin SVM

- Problem: Given a set $\mathcal{S}$ of points $x_i \in \mathcal{S}$ with labels $y_i \in \{-1, 1\}$
- Solution:

$$\begin{array}{ll} \max & \gamma \\ \text{subject to} & y_i[w \cdot x_i + b] \geq \gamma \quad \forall i \\ & \|w\| = 1 \end{array}$$

- Note:
 - If $y_i = 1$: $w \cdot x_i \geq \gamma$
 - If $y_i = -1$: $w \cdot x_i \leq -\gamma$



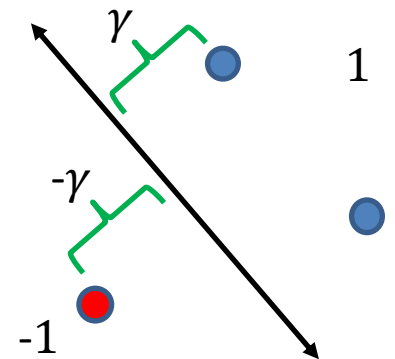
Hard-margin SVM

- Problem: Given a set $\mathcal{S}$ of points $x_i \in \mathcal{S}$ with labels $y_i \in \{-1, 1\}$
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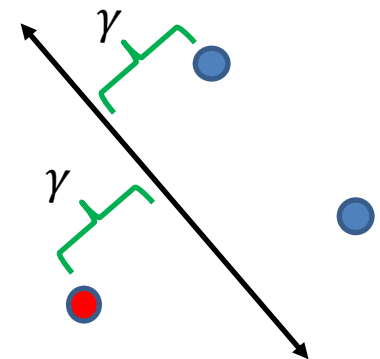
b can be left out.
Then hyperplane goes through the origin

- Note:
 - If $y_i = 1$: $w \cdot x_i \geq \gamma$
 - If $y_i = -1$: $w \cdot x_i \leq -\gamma$



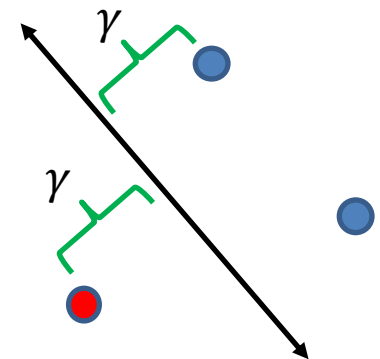
Hard-margin SVM

- There's a different way to represent the same program
- If $\mathbf{w}$ is normalized in the program (and points x_i have norm at most 1)
 - Equation $y_i[w \cdot x_i + b] \geq \gamma$ has meaning
 - Margin γ is relative to radius 1



Hard-margin SVM

- If $\mathbf{w}$ is not normalized
 - Some other measure needs to decide what relative distances are
 - This will be the max margin: Take $\gamma = 1$
- Now the goal becomes
 - Maximize $\min_{x_i} \frac{|w \cdot x_i + b|}{\|w\|}$
 - Subject to $|w \cdot x_i + b| \geq 1$
 - So maximize $\frac{1}{\|w\|}$
 - Or minimize $\|w\|$

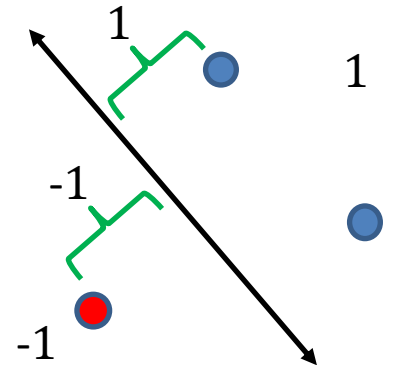


Hard-margin SVM

- Problem: Given a set $\mathbf{S}$ of points $x_i \in \mathbf{S}$ with labels $y_i \in \{-1, 1\}$
- Solution: If points x_i are **not** normalized

$$\begin{array}{ll} \min & \|w\| \\ \text{subject to} & y_i[w \cdot x_i + b] \geq 1 \quad \forall i \end{array}$$

- Can be solved by convex optimization
 - Very expensive!



Margin with error

- Now suppose the set isn't separable
 - Every hyperplane has some errors
- We know that with probability at least $1 - \delta$

$$e(h) \leq \bar{e}(h) + \sqrt{\frac{8d \ln \frac{2en}{d} + 8 \ln \frac{4}{\delta}}{n}}$$

Margin with error

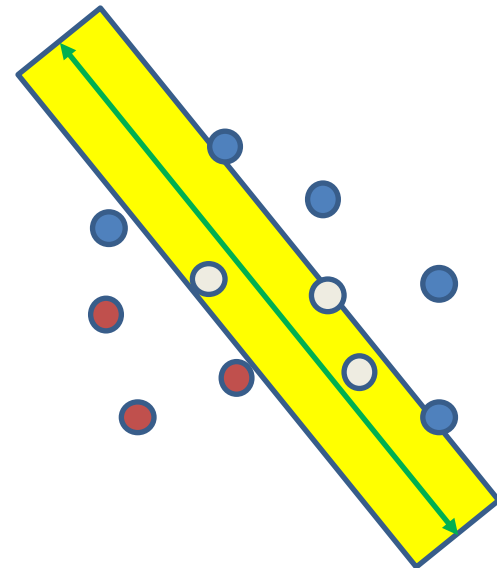
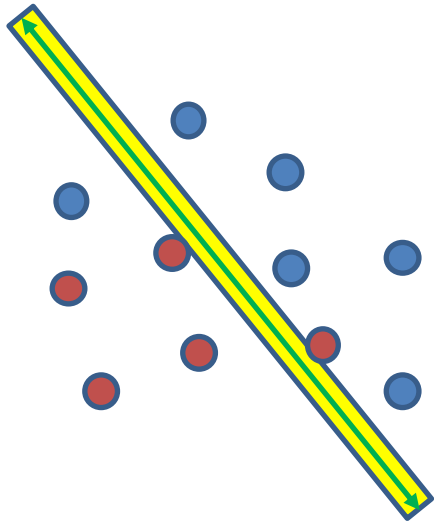
- We know that with probability at least $1 - \delta$

$$e(h) \leq \bar{e}(h) + \sqrt{\frac{8d \ln \frac{2en}{d} + 8 \ln \frac{4}{\delta}}{n}}$$

- Tradeoff:
 - Larger margin $\rightarrow$ smaller dimension $d = \frac{O(1)}{\gamma_2}$
 - Smaller margin $\rightarrow$ smaller empirical error $\bar{e}(h)$

Margin with error

- For the same hyperplane, we can consider different margins
 - The wider margin has higher empirical error (all points falling in the margin)
 - but lower VC-dimension

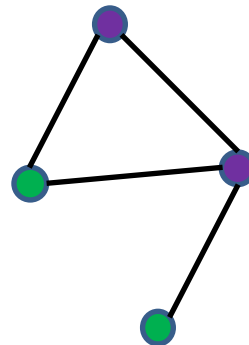
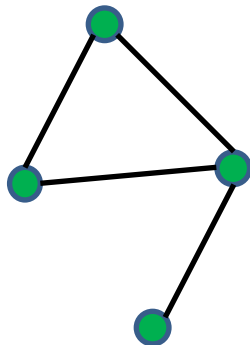


Minimizing errors

- Let **S** be a set of n d -dimensional points
- Problem:
 - Given **S**
 - Find a $(d-1)$ -dimensional hyperplane with as few errors as possible.
- Claim: NP-hard

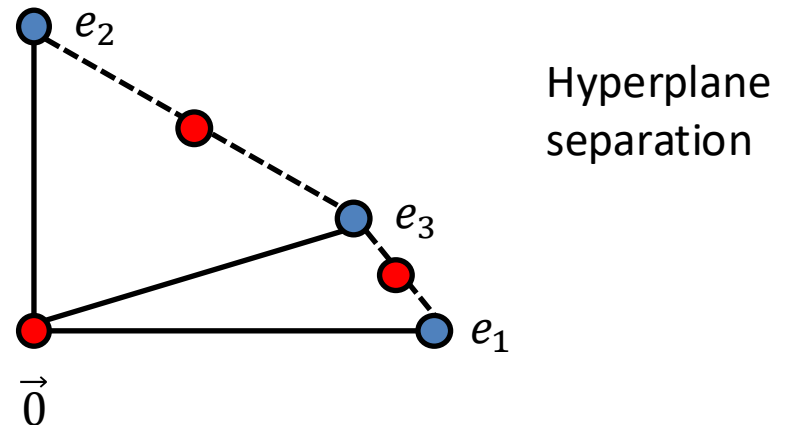
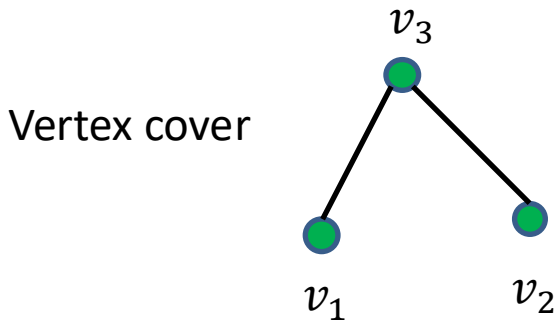
Minimizing errors

- Proof: Reduction from Min Vertex Cover
- Vertex cover:
 - Input: a graph $G=(V,E)$ of vertices V and edges E
 - Output: the smallest subset of V which *covers* all edges
 - Classic NP-hard problem



Minimizing errors

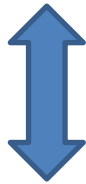
- Reduction: Given an instance $G=(V,E)$ of Vertex Cover
- Create an instance S of min hyperplane errors problem:
 - Origin is labelled positive with weight n
 - Each vertex $v_i \in V$ corresponds to basis vector e_i , and is labelled negative.
 - If two vertices $v_i, v_j \in V$ are connected, add a positive point with weight n in the middle of e_i, e_j



Minimizing errors

- Claim: In the reduction, the following are equivalent:

- a hyperplane erring on $k < n$ points in P

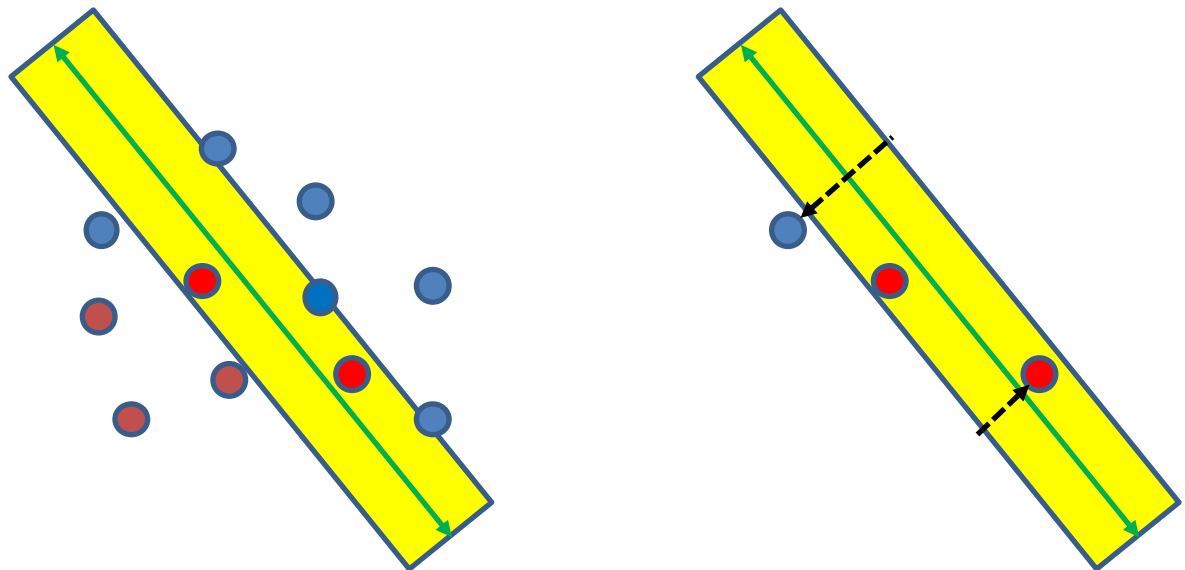


- A vertex cover for G of size k

- Back to SVM...

Support Vector Machines

- Vapnik and Cortes considered SVM when the point set isn't separable.
- Idea: **Soft margin** – errors allowed
 - For every point x_i on the wrong side
 - pay a penalty ζ_i equal to how far it's on the wrong side
 - Want to minimize the sum of penalties



Soft margin SVM

- The convex program is:

- Min $\frac{1}{n} \sum_i \zeta_i + \lambda \|w\|^2$

- Subject to

$$y_i(w \cdot x_i - b) \geq 1 - \zeta_i \quad \forall i$$

$$\zeta_i \geq 0 \quad \forall i$$

- $\frac{1}{n} \sum_i \zeta_i$ Is the **hinge loss**.

- What is λ ?

- Trying to minimize two things at once!
 - Gives trade-off between them

Soft margin SVM

- Who chooses λ ?
 - Algorithm, uses **cross-validation**
- Cross validation: Technique for choosing algorithm parameters
 - Training set is divided into two:
 - **Training** and **validation**
 - Run SVM on training with different values of λ
 - Test all runs on validation set, and choose best λ

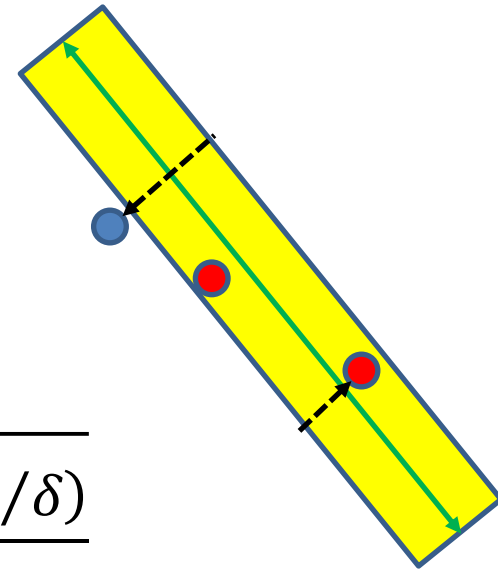
Generalization bounds

- What is the true error of SVM?
- Hinge loss $\geq$ empirical error
 - $\zeta_i \geq \gamma[h(x_i) \neq y_i] \geq [h(x_i) \neq y_i]$
- Hoeffding-like result:
 - Let w be the output of SVM. Then with probability at least $1 - \delta$

$$e(h) \leq \sum_i \zeta_i + \frac{4\|w\|}{\sqrt{n}} + (1 + 2\|w\|) \sqrt{\frac{2 \ln(4\|w\|/\delta)}{n}}$$

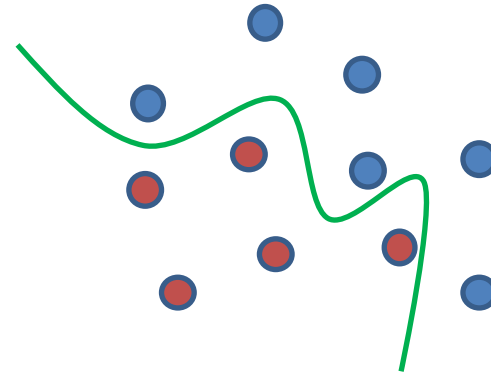
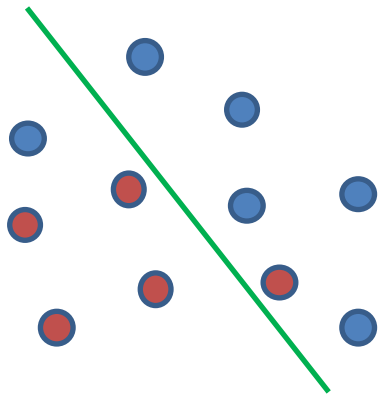


Bounds
Empirical
error



Non-linear SVM

- What if we want a separator that's not a hyperplane?
 - Not linear.



Non-linear SVM

- Soft margin SVM can be rewritten as the dual:

$$\min \sum_{i=1}^n c_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i c_i (x_i \cdot x_j) y_j c_j$$

subject to

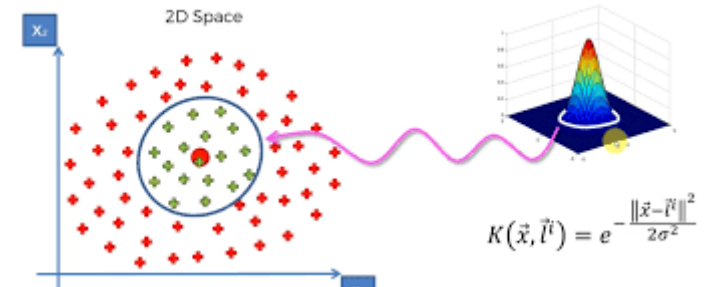
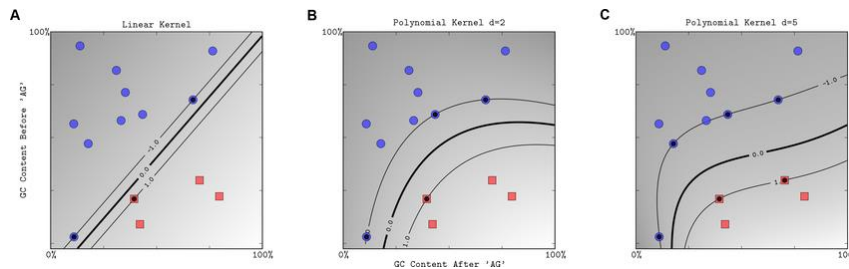
$$\sum_{i=1}^n c_i y_i = 0$$

$$0 \leq c_i \leq \frac{1}{2n\lambda}$$

- Dot product $(x_i \cdot x_j)$ is known as a **kernel**

Non-linear SVM

- Replace $(x_i \cdot x_j)$ by other functions. Popular choices:
 - Polynomial kernel: $(x_i \cdot x_j + 1)^d$
 - Radial Basis Function (RBF): $e^{-\|x_i - x_j\|/2\sigma^2}$



- What does this do?
 - embeds points in a high dimensional space
 - Finds a linear separator in high dimension
 - Corresponds to a non-linear separator in original space